EXACT FUNCTORS IN THE CORRESPONDENCE HAVE KERNELS FLAT OVER X

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0FZD: sources/stacks-chow/tags.txt, source lines 19948–19948. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GPA: sources/stacks-chow/tags.txt, source lines 20820–20820. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0FZF: sources/stacks-chow/tags.txt, source lines 19950–19950. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0GNX: sources/stacks-chow/tags.txt, source lines 20807–20807. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Lemma 0FZD or in Lemma 0GPA if F is an exact functor, then the corresponding object $\mathcal{K}$ of $QCoh(\mathcal{O}_{X \times_R Y})$ is flat over X.*

Proof. We may assume X is affine, so we are in the case of Lemma 0FZD. By Lemma 0FZF we may assume Y is affine. In the affine case the statement translates into Remark 0GNX. □

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